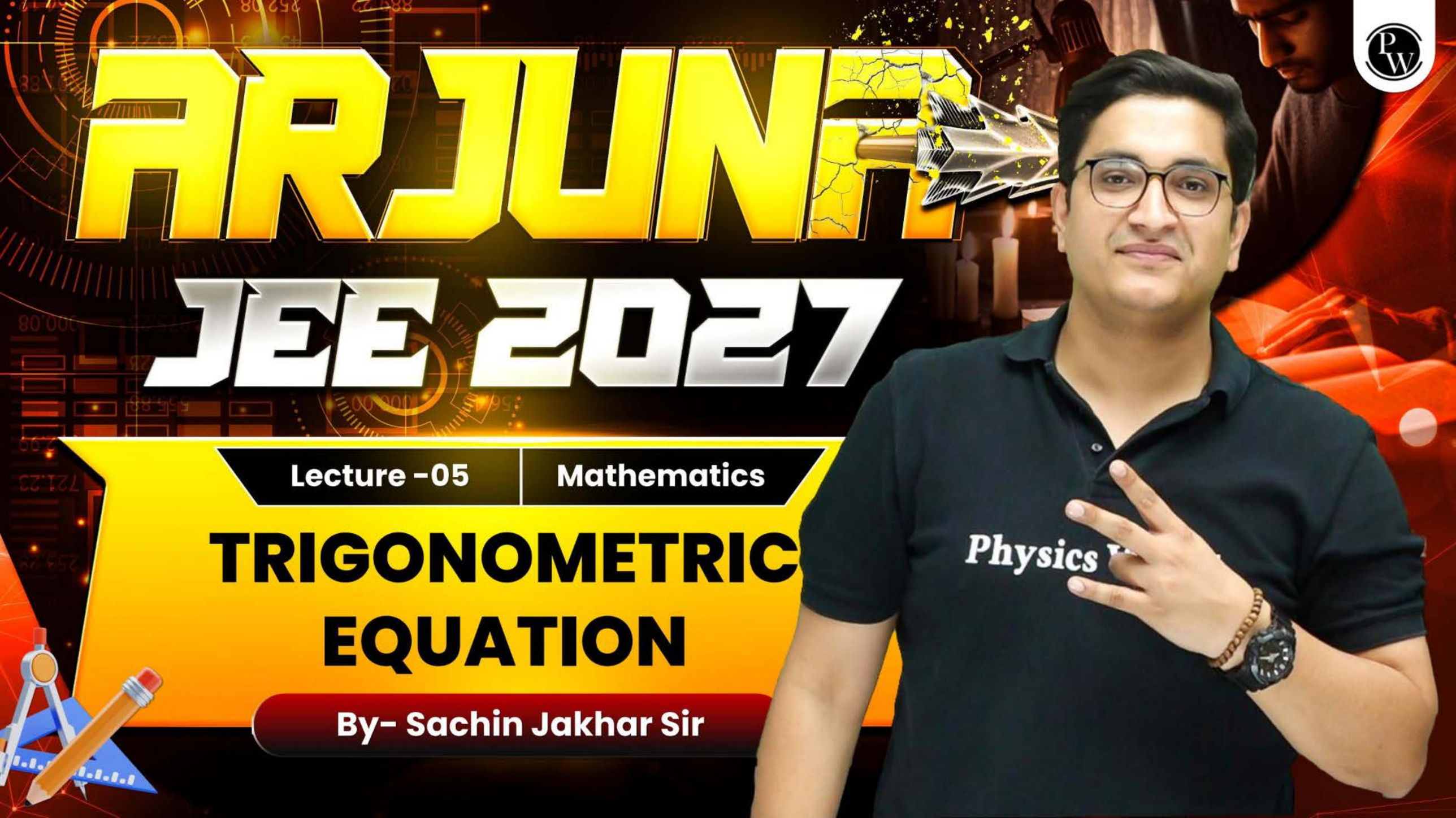
ARJUNA JEE 2027

Lecture -05 | Mathematics

TRIGONOMETRIC EQUATION

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Topics *to be covered*

A **Types of Trigonometric Equations**

B **Question Practice**

C

D



H.W. Discussion

Challenger Question

Principal solution of equation : $\sin^2 x + 2\tan^2 x + \frac{4}{\sqrt{3}}\tan x - \sin x + \frac{11}{12} = 0$

$$\underbrace{\sin^2 x - 2\cdot\frac{1}{2}\sin x + \frac{1}{4}} - \frac{1}{4} + 2\left\{\underbrace{\tan^2 x + 2\cdot\frac{1}{\sqrt{3}}\tan x + \frac{1}{3}} - \frac{1}{3}\right\} + \frac{11}{12} = 0$$

$$\left(\sin x - \frac{1}{2}\right)^2 - \frac{1}{4} + 2\left(\tan x + \frac{1}{\sqrt{3}}\right)^2 - \frac{2}{3} + \frac{11}{12} = 0$$

$$\underbrace{\left(\sin x - \frac{1}{2}\right)^2}_{\geq 0} + 2\underbrace{\left(\tan x + \frac{1}{\sqrt{3}}\right)^2}_{\geq 0} - \cancel{\frac{11}{12}} + \cancel{\frac{11}{12}} = 0$$

$\sin x = \pm\frac{1}{2}$ and $\tan x = \mp\frac{1}{\sqrt{3}}$ $\Rightarrow x = \pi - \frac{\pi}{6} = \boxed{\frac{5\pi}{6}}$

General soln.

$x \in \left\{2n\pi + \frac{5\pi}{6}\right\}, n \in I.$

Question [JEE (Adv.)-2018]

Let a, b, c be three non zero numbers such that the equation $\sqrt{3}a\cos x + 2b\sin x = c,\ x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then the value of $\frac{b}{a}$ is

$\sqrt{3}a\cos x + 2b\sin x = c$ (α, β)

$\sqrt{3}a\cos\alpha + 2b\sin\alpha = c$ —(1)

$\ominus$ $\sqrt{3}a\cos\beta + 2b\sin\beta = c$ —(2)

$\sqrt{3}a(\underbrace{\cos\alpha - \cos\beta}_{C\ \ D}) + 2b(\sin\alpha - \sin\beta) = 0$

$\sqrt{3}a\left(-2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}\right) + 2b\left(2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}\right) = 0$ ($\frac{\alpha+\beta}{2} = 60°$)

$\sin\frac{\alpha-\beta}{2}\left\{-2\sqrt{3}a\sin 30° + 4b\cos 30°\right\} = 0$

$-2\sqrt{3}a \times \frac{1}{2} + 4b \times \frac{\sqrt{3}}{2} = 0$

$2b\sqrt{3} = \sqrt{3}a$

$\frac{b}{a} = \frac{1}{2} = 0.5$

Question [JEE Mains-2021]

The number of roots of the equation $(81)^{\sin^2 x} + (81)^{\cos^2 x} = 30$ in the interval $[0, \pi]$ is equal to:

A 3

B 4

C 8

D 2

Say: $81^{\sin^2 x} = t$

$\Rightarrow 81^{\cos^2 x} = 81^{1-\sin^2 x}$

$= \frac{81^1}{81^{\sin^2 x}} = \frac{81}{t}$

$t + \frac{81}{t} = 30$

$\Rightarrow \frac{t^2 + 81}{t} = 30$

$\Rightarrow t^2 - 30t + 81 = 0$

$\Rightarrow (t-27)(t-3) = 0$

$t = 27$ or 3

$t = 81^{\sin^2 x} = 27$

$\left(3^4\right)^{\sin^2 x} = 3^3$

$4\sin^2 x = 3$

$\sin^2 x = \frac{3}{4}$

$\sin x = \pm\frac{\sqrt{3}}{2}$

OR

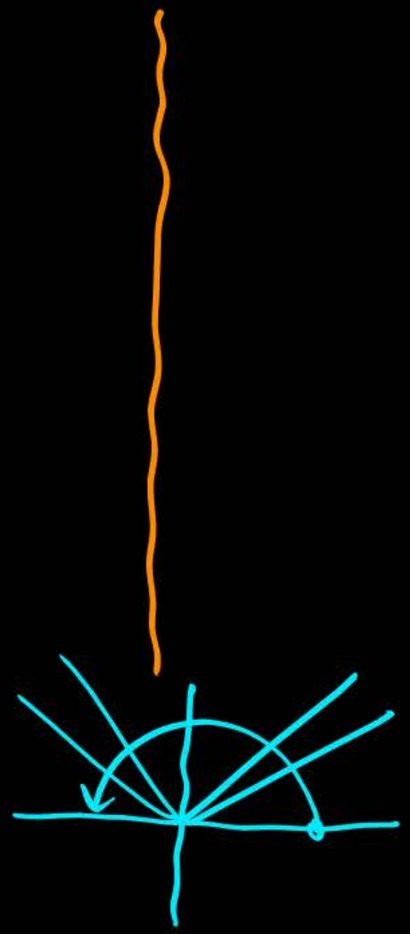
$t = 81^{\sin^2 x} = 3$

$\left(3^4\right)^{\sin^2 x} = 3^1$

$4\sin^2 x = 1$

$\sin^2 x = \frac{1}{4}$

$\sin x = \pm\frac{1}{2}$



Question [JEE Mains-2023]

Let $S = \left\{x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) : 9^{1-\tan^2 x} + 9^{\tan^2 x} = 10\right\}$ and $\beta = \sum_{x \in S} \tan^2\left(\frac{x}{3}\right)$

then $\frac{1}{6}(\beta - 14)^2$ is equal to

A 16

B 32

C 8

D 64

$\frac{1}{6}\left(-8\sqrt{3}\right)^2 = \frac{1}{6} \times 64 \times 3 = 32$

Question [JEE Mains-2023]

Let $S = \left\{x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) : 9^{1-\tan^2 x} + 9^{\tan^2 x} = 10\right\}$ and $\beta = \sum_{x \in S} \tan^2\left(\frac{x}{3}\right)$

$9^{\tan^2 x} = t$

$9^{1-\tan^2 x} = \frac{9^1}{9^{\tan^2 x}} = \frac{9}{t}$

$S = \left\{\frac{\pi}{4}, -\frac{\pi}{4}, 0\right\}$

$\frac{9}{t} + t = 10$

$9 + t^2 = 10t$

$t^2 - 10t + 9 = 0$

$t^2 - 9t - t + 9 = 0$

$t(t-9) - 1(t-9) = 0$

$(t-1)(t-9) = 0$

$t = 1$ OR $t = 9$

$9^{\tan^2 x} = 9^0$

$\tan^2 x = 0$

$\tan x = 0$

$9^{\tan^2 x} = 9^1$

$\tan^2 x = 1$

$\tan x = \pm 1$

$\beta = \tan^2 \frac{\pi}{12} + \tan^2\left(-\frac{\pi}{12}\right) + \tan^2 0$ ($\frac{\pi}{12} = 15^\circ$, $\tan^2 0 = 0$)

$\beta = (2-\sqrt{3})^2 + (2-\sqrt{3})^2$

$= 2(2-\sqrt{3})^2$

$= 2(4+3-4\sqrt{3})$

$\beta = 14 - 8\sqrt{3} \Rightarrow \beta - 14 = -8\sqrt{3}$

Question [JEE Mains-2022]

The number of solutions of the equation

$$\cos\left(x+\frac{\pi}{3}\right)\cos\left(\frac{\pi}{3}-x\right)=\frac{1}{4}\cos^2 2x, x\in[-3\pi,3\pi]$$ **is:**

A. 8

B. 5

C. 6

D. 7

$$\cos^2\frac{\pi}{3}-\sin^2 x=\frac{1}{4}\cos^2 2x.$$

$$\frac{1}{4}-\sin^2 x=\frac{1}{4}\cos^2 2x.$$

$$\frac{1}{4}(1-\cos^2 2x)=\sin^2 x.$$

$$\frac{\sin^2 2x}{4}=\sin^2 x$$

$$\sin^2 x\cdot\cos^2 x-\sin^2 x=0$$

$$\sin^2 x\cdot(\cos^2 x-1)=0$$

$$\sin^2 x\cdot(-\sin^2 x)=0$$

$$\sin^4 x=0$$

$$\sin x=0.$$

$$x=-3\pi,-2\pi,-\pi,0,\pi,2\pi,3\pi.$$

Question

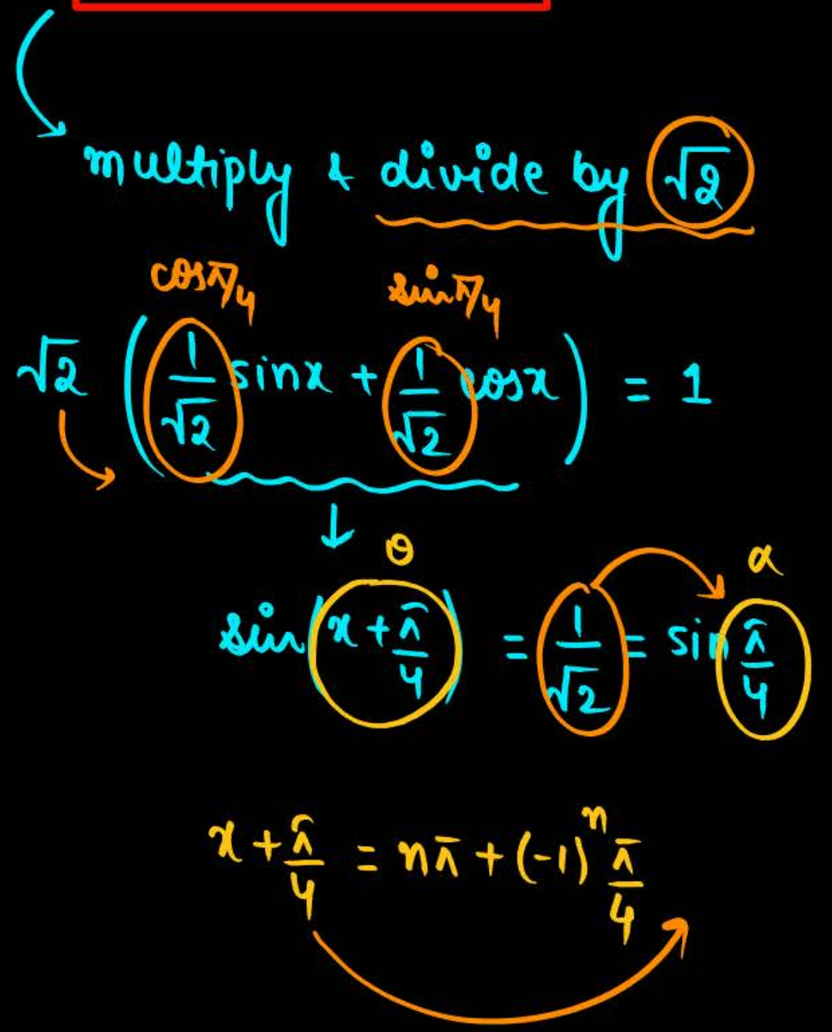
The general solution of the trigonometric equation $\sin x + \cos x = 1$ is given by

A $x = 2n\pi; n = 0, \pm 1, \pm 2, \ldots$

B $x = 2n\pi + \pi/2; n = 0, \pm 1, \pm 2, \ldots$

C $x = n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}; n = 0, \pm 1, \pm 2, \ldots$

D None of the above





Find the number of solution of the equation in $[0, 2\pi]$,

$\tan(5\pi\cos\alpha) = \cot(5\pi\sin\alpha)$.

θ_1 θ_2

$\tan\theta_1 = \cot\theta_2$

$\tan\theta_1 = \tan\left(\frac{\pi}{2} - \theta_2\right)$

$\theta_1 = n\pi + \frac{\pi}{2} - \theta_2$

$\theta_1 + \theta_2 = n\pi + \frac{\pi}{2} = \frac{2n\pi + \pi}{2} = (2n+1)\frac{\pi}{2}$, $n \in I$.

$5\pi\cos\alpha + 5\pi\sin\alpha = (2n+1)\frac{\pi}{2}$

$\cos\alpha + \sin\alpha = \frac{(2n+1)}{10}$, $n \in I$.

$n_f = 14 \times 2 = 28$ sol^n.

$\cos\alpha + \sin\alpha = \dfrac{(2n+1)}{10}$, $n \in I$.

Range $\equiv [-\sqrt{2}, \sqrt{2}]$

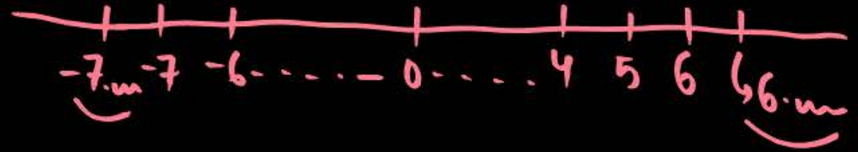
$$-\sqrt{2} \le \frac{2n+1}{10} \le \sqrt{2}$$

$$-10\sqrt{2} \le 2n+1 \le 10\sqrt{2}$$

$$\frac{-10\sqrt{2}-1}{2} \le n \le \frac{10\sqrt{2}-1}{2}$$

$\frac{-10\sqrt{2}-1}{2}$ $\rightarrow$ $-7.\ldots$

$\frac{10\sqrt{2}-1}{2}$ $\rightarrow$ $6.\ldots$



n = 6, 5, 4, 3, 2, 1, 0, −1, −2, −3, −4, −5, −6, −7.

14 values.

$\alpha \in [0, 2\pi]$

if $n = 1$

$\cos\alpha + \sin\alpha = \dfrac{3}{10}$

$\alpha \in [0, 2\pi]$

if $n = 1$

$$\cos\alpha + \sin\alpha = \frac{3}{10} \longrightarrow \text{nos} = (2)$$

$\times \& \div \sqrt{2}$

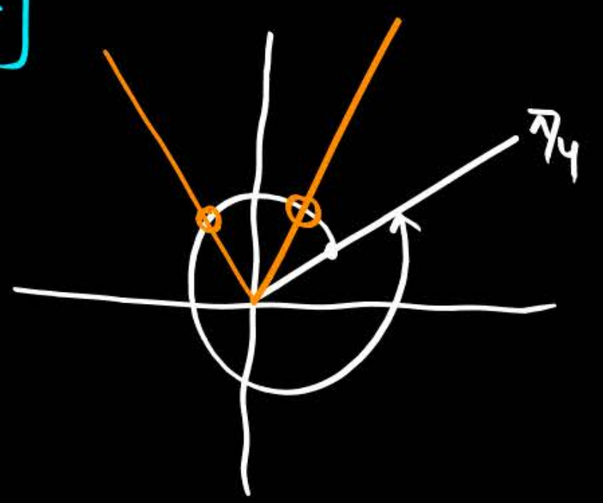
$$\sin\left(\underbrace{\alpha + \frac{\pi}{4}}_{\theta}\right) = \frac{3}{10\sqrt{2}}$$

$\alpha \in [0, 2\pi]$

$\alpha + \frac{\pi}{4} \in \left[\frac{\pi}{4}, \frac{9\pi}{4}\right]$

$$\sin\theta = \frac{3}{10\sqrt{2}}, \quad \theta \in \left[\frac{\pi}{4}, \frac{9\pi}{4}\right]$$

nos = (2)



Question [JEE Mains-2023]

Let $S = \{\theta \in [0, 2\pi) : \tan(\pi\cos\theta) + \tan(\pi\sin\theta) = 0\}$. Then $\sum_{\theta \in S} \sin^2\left(\theta + \frac{\pi}{4}\right)$ is equal to



TYPE-03

Solving equations by transforming a sum of trigonometric functions into a product.

Question [JEE Mains-2021]

The sum of all values of x in $[0, 2\pi]$, for which $\sin x + \sin 2x + \sin 3x + \sin 4x = 0$, is equal to

A 11π

B 9π

C 8π

D 12π

$$2\sin\frac{3x}{2}\cos\frac{x}{2} + 2\sin\frac{7x}{2}\cos\frac{x}{2} = 0$$

$$2\cos\frac{x}{2}\left\{\sin\frac{3x}{2} + \sin\frac{7x}{2}\right\} = 0$$

$$2\cos\frac{x}{2} \cdot 2\sin\frac{5x}{2}\cos x = 0$$

$$\cos\frac{x}{2} = 0 \quad \text{OR} \quad \sin\frac{5x}{2} = 0 \quad \text{OR} \quad \cos x = 0$$

$$\frac{x}{2} = (2n+1)\frac{\pi}{2} \qquad \frac{5x}{2} = n\pi \qquad x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$x = \pi \qquad x = \frac{2n\pi}{5}$$

$$x = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}, 2\pi$$

$$\text{Sum} = \pi + 2\pi + 2\pi + 4\pi = 9\pi$$

Question

General solution of $\sin x + \sin 5x = \sin 2x + \sin 4x$ is

A $\frac{n\pi}{3}$

B $\frac{2n\pi}{3}$

C $2n\pi$

D $n\pi$

TYPE-04

Solving equations by transforming a product of trigonometric functions into a sum.

Question

Solve: $\cos x \cdot \cos 2x \cdot \cos 3x = \frac{1}{4}$

$$\frac{1}{2}\left(2\cos x \cdot \cos 2x\right)\cos 3x = \frac{1}{4}$$

$$\left(\cos(x) + \cos 3x\right)\cos 3x = \frac{1}{2}$$

$$2\cos x \cdot \cos 3x + 2\cos^2 3x = 1$$

$$\cos(2x) + \cos 4x + 2\cos^2 3x - 1 = 0$$

$$\cos 2x + \cos 4x + \cos 6x = 0$$

$$2\cos 4x \cdot \cos 2x + \cos 4x = 0$$

$$\cos 4x \cdot (2\cos 2x + 1) = 0$$

$\cos 4x \cdot (2\cos 2x + 1) = 0$

$\cos 4x = 0$ OR $\cos(2x) = -\frac{1}{2} = \cos\frac{2\pi}{3}$

θ and α

$4x = (2n+1)\frac{\pi}{2}$

$x = (2n+1)\frac{\pi}{8}$

$2x = 2n\pi \pm \frac{2\pi}{3}$

$x = \frac{3n\pi \pm \pi}{3}$

$x = (3n \pm 1)\frac{\pi}{3}$

$\cup$

Question

Number of solutions of the trigonometric equation in $[0, \pi]$, is $\sin 3\theta = 4\sin\theta \cdot \sin 2\theta \cdot \sin 4\theta$.



A 4

B 16

C 8

D 0

Question [JEE Mains-2023]

If m and n respectively are the numbers of positive and negative value of θ in the interval $[-\pi, \pi]$ that satisfy the equation

$2\cos 2\theta \cos\frac{\theta}{2} = 2\cos 3\theta \cos\frac{9\theta}{2}$, then mn is equal to

$mn = 25$ Ans.

$\cos\left(\frac{3\theta}{2}\right) + \cos\left(\frac{5\theta}{2}\right) = \cos\frac{3\theta}{2} + \cos\frac{15\theta}{2}$

$\cos\frac{15\theta}{2} = \cos\frac{5\theta}{2}$

$\frac{15\theta}{2} = 2n\pi \pm \frac{5\theta}{2}$

$\frac{15\theta}{2} = \frac{4n\pi \pm 5\theta}{2}$

$$15\theta = 4n\pi \pm 5\theta$$

$\oplus$ $\qquad$ $\ominus$

$15\theta = 4n\pi + 5\theta$ $\qquad$ OR $\qquad$ $15\theta = 4n\pi - 5\theta$

$10\theta = 4n\pi$ $\qquad$ $20\theta = 4n\pi$

$\theta = \frac{4n\pi}{10} = \frac{2n\pi}{5}$. $\qquad$ $\theta = \frac{4n\pi}{20}$

$\theta = \frac{n\pi}{5}$.

$\theta \in [-\pi, \pi]$

$n = 5, \; n = 5$

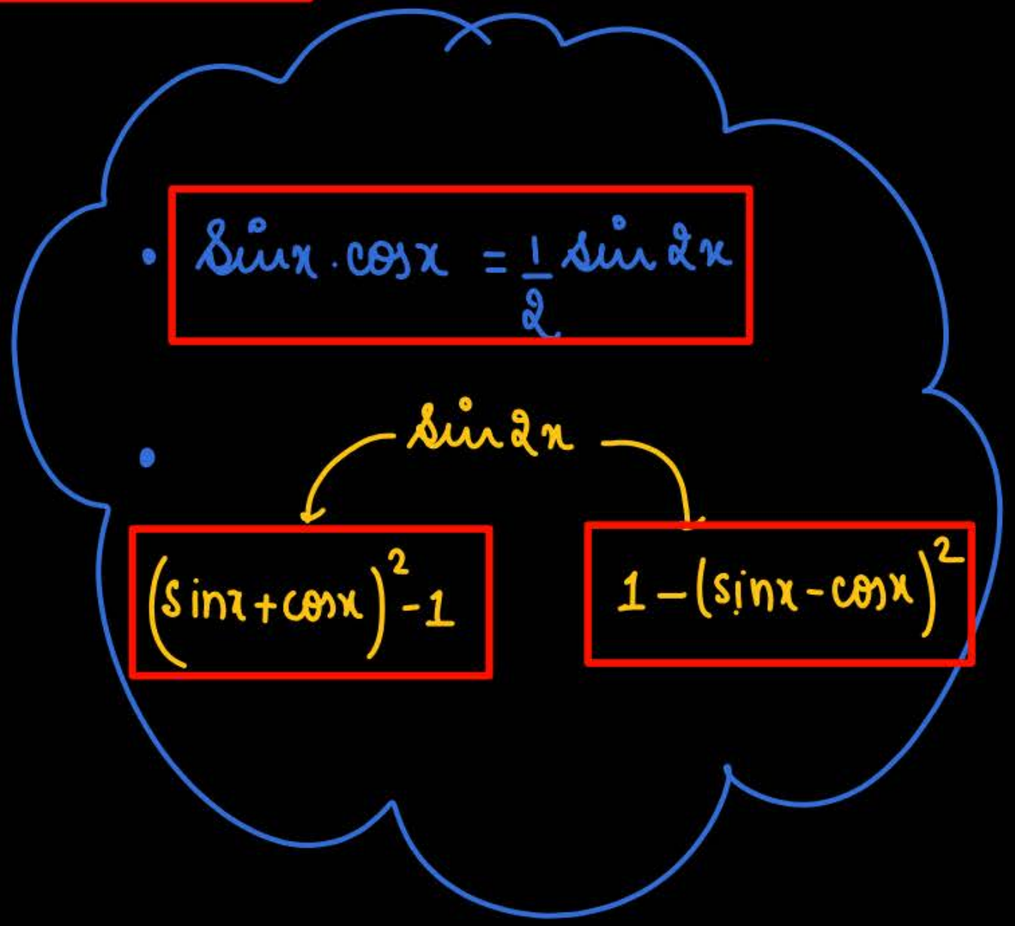
$0, \frac{2\pi}{5}, \frac{4\pi}{5}, -\frac{2\pi}{5}, -\frac{4\pi}{5}$.

$0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}, \pi, -\frac{\pi}{5}, -\frac{2\pi}{5}, -\frac{3\pi}{5}, -\frac{4\pi}{5}, -\pi$.

TYPE-05

Solving equations by a change of variable or by substitution method:
Equation of the form $f(\sin x \pm \cos x, \sin x \cdot \cos x) = 0$

Put $\sin x \pm \cos x = t$



Question

Solve the equation : $\sin x + \cos x - 2\sqrt{2}\sin x \cos x = 0$.

put $\sin x + \cos x = t$.

SQS.

$1 + \sin 2x = t^2$

$\sin 2x = t^2 - 1$

$\sin x + \cos x - \sqrt{2}(\sin 2x) = 0$.

$t - \sqrt{2}(t^2 - 1) = 0$.

$t - \sqrt{2}t^2 + \sqrt{2} = 0$

$\sqrt{2}t^2 - t - \sqrt{2} = 0$.

$\sqrt{2}t^2 - 2t + t - \sqrt{2} = 0$

$\sqrt{2}t(t - \sqrt{2}) + 1(t - \sqrt{2}) = 0$

$(\sqrt{2}t + 1)(t - \sqrt{2}) = 0$

$t = \sqrt{2}$ OR $-\frac{1}{\sqrt{2}}$.

$\sin x + \cos x = \sqrt{2}$ OR $-\frac{1}{\sqrt{2}}$

QIBY !!

Question

Solve the equation : $\sin x + \cos x - 2\sqrt{2}\sin x \cos x = 0$.

M-2

$$\sqrt{2}\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x\right) = \sqrt{2}\sin 2x$$

$$\sin\left(x + \frac{\pi}{4}\right) = \sin 2x$$

(θ and α)

$$x + \frac{\pi}{4} = n\pi + (-1)^n 2x$$

$n = 2m+1$ OR $n = 2m$ $\cup$.

Question

Solve: $\sin^2 x + (\sqrt{3} - 1)\sin x \cos x - \sqrt{3}\cos^2 x = 0.$

$$\sin^2 x + \sqrt{3}\sin x\cos x - \sin x\cos x - \sqrt{3}\cos^2 x = 0.$$

$$\sin x(\sin x - \cos x) + \sqrt{3}\cos x(\sin x - \cos x) = 0.$$

$$(\sin x - \cos x)\{\sin x + \sqrt{3}\cos x\} = 0.$$

$\sin x = \cos x$ OR $\sin x = -\sqrt{3}\cos x$

$\tan x = 1$ OR $\tan x = -\sqrt{3}$ ✓

TYPE-06 (A)

Solving equations with the use of boundedness of the function (or)

By comparing range of both side.

RAPID FIRE

Solve for x :

1. $\sin x = 5$ ⟶ ϕ.
2. $3\sin x - 4\cos x = 7$ ⟶ ϕ.

Range of RHS $\in [-5, 5]$

3. $\sin x + \cos x = \sqrt{2}$
4. $\cos x + \cos 2x + \cos 3x = 3$

≤ 1 ≤ 1 ≤ 1

RHS ki max. value = 3.

RHS $[-\sqrt{2}, \sqrt{2}]$

equality hold : $\cos x = 1 \cap \cos 2x = 1 \cap \cos 3x = 1$.

$x = n\pi$ $\quad 2x = n\pi \Rightarrow x = \frac{n\pi}{2}$ $\quad 3x = n\pi \Rightarrow x = \frac{n\pi}{3}$

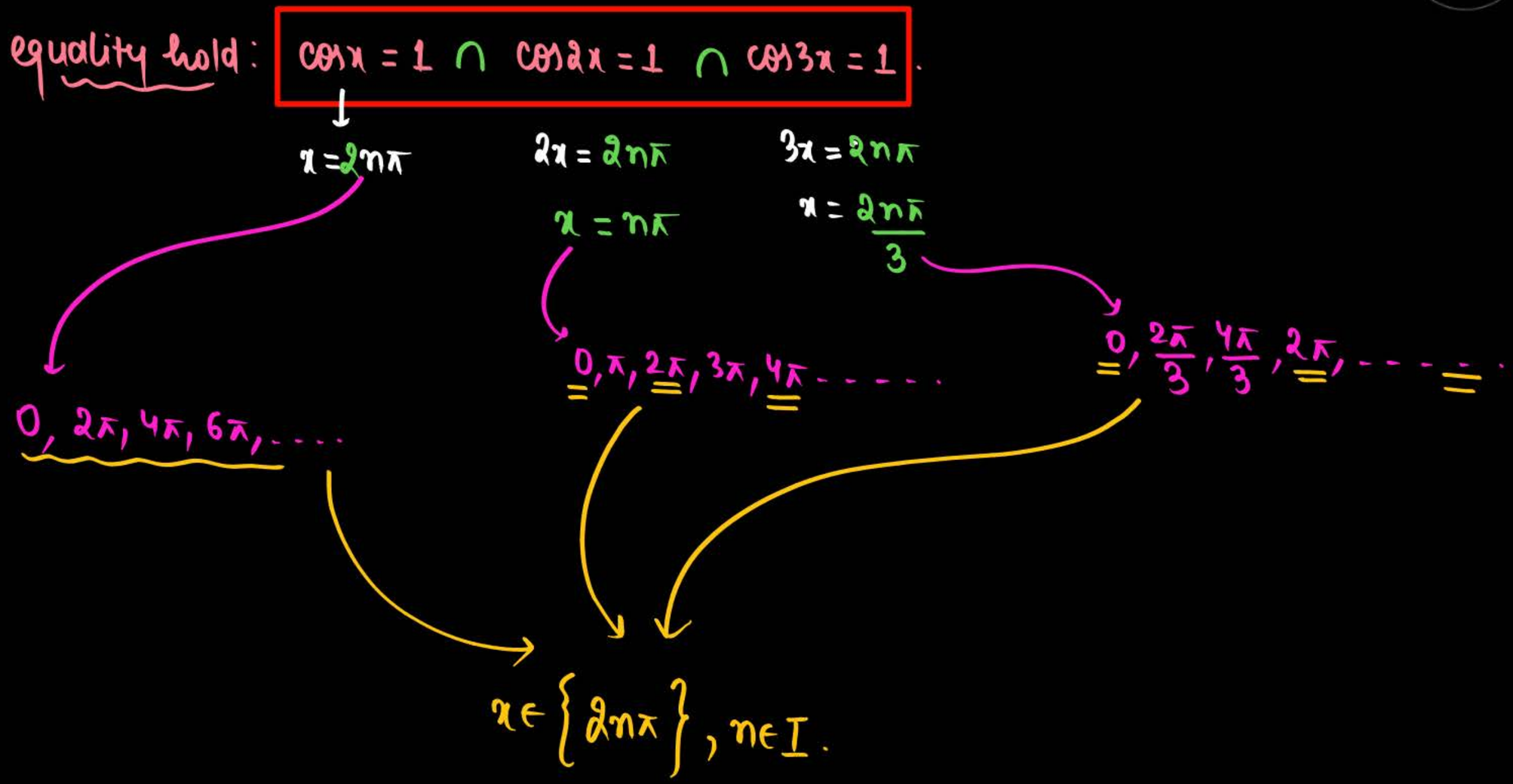
(iii)

$\sin x + \cos x = \sqrt{2}$ OR $\sin\left(x + \frac{\pi}{4}\right) = 1$

$x + \frac{\pi}{4} = (4n+1)\frac{\pi}{2}$ ✓

equality hold $\sin x = \cos x = \frac{1}{\sqrt{2}}$




equality hold : $\cos x = 1 \cap \cos 2x = 1 \cap \cos 3x = 1$.
$x = 2n\pi$
$2x = 2n\pi$
$x = n\pi$
$3x = 2n\pi$
$x = \frac{2n\pi}{3}$
$0, 2\pi, 4\pi, 6\pi, \ldots$
$0, \pi, 2\pi, 3\pi, 4\pi \ldots\ldots$
$0, \frac{2\pi}{3}, \frac{4\pi}{3}, 2\pi, \ldots\ldots$
$x \in \{2n\pi\}, n \in I$.

Question [JEE Mains-2024]

The number of solutions of the equation $4\sin^2 x - 4\cos^3 x + 9 - 4\cos x = 0; x \in [-2\pi, 2\pi]$ is

A. 1

B. 3

C. 2

D. 0

$4(1-\cos^2 x) - 4\cos^3 x + 9 - 4\cos x = 0.$

$4 - 4\cos^2 x - 4\cos^3 x + 9 - 4\cos x = 0$

$13 = 4\cos^2 x + 4\cos^3 x + 4\cos x$

max. max. max.

$4 \quad 4 \quad 4$

$\text{Max.} = 12.$

no soln.

Note :-

if an eqⁿ. is Consistent that means eqⁿ has solⁿ.

RAPID FIRE ROUND

Find value of k for which following equation has atleast one solution (or consistent):

(i) $12\cos x - 5\sin x = k$

(ii) $\sin x \cdot \cos x = k$

(iii) $\sin x + 1 = k\sin x + 2(k-1)$

(i)

$k \in [-13, 13]$

(ii)

$\frac{\sin 2x}{2} = k$

$\sin 2x \in [-1, 1]$

$\frac{\sin 2x}{2} \in \left[-\frac{1}{2}, \frac{1}{2}\right]$

$k \in \left[-\frac{1}{2}, \frac{1}{2}\right]$

RAPID FIRE ROUND

Find value of k for which following equation has atleast one solution (or consistent):

(i) $12\cos x - 5\sin x = k$

(ii) $\sin x \cdot \cos x = k$

(iii) $\sin x + 1 = k\sin x + 2(k-1)$

$\sin x + 1 = k\sin x + 2k - 2$

$\sin x + 3 = k(\sin x + 2)$

$\frac{\sin x + 3}{\sin x + 2} = k$

$\frac{\sin x + 2}{\sin x + 2} + \frac{1}{\sin x + 2} = k$

$\# \quad 1 + \frac{1}{\sin x + 2} = k, \; k \in \left[\frac{4}{3}, 2\right]$

$\sin x \in [-1, 1]$

$\sin x + 2 \in [1, 3]$

$\frac{1}{\sin x + 2} \in \left[\frac{1}{3}, 1\right]$ $\xrightarrow{+1}$ $1 + \frac{1}{\sin x + 2} \in \left[\frac{4}{3}, 2\right]$

Question [JEE Mains-2023]

The set of all values of λ for which the equation $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$ has a real solution x, is

A [−2, −1]

B [−3/2, −1]

C [−2, −3/2]

D [−1, −1/2]

$$\text{LHS} = (2\cos^2 x - 1)^2 - 2(1-\cos^2 x)^2 - 2\cos^2 x.$$

$\cos^2 x = t$

$$\text{LHS} = (2t-1)^2 - 2(1-t)^2 - 2t$$

$$= 4t^2 - 4t + 1 - 2(1 + t^2 - 2t) - 2t$$

$$= 4t^2 - 4t + 1 - 2 - 2t^2 + 4t - 2t$$

$$= 2t^2 - 2t - 1.$$

$$2t(t-1) - 1.$$

$$2\cos^2 x(\cos^2 x - 1) - 1.$$

$$2\cos^2 x(-\sin^2 x) - 1$$

$$-2(\sin x \cos x)^2 - 1.$$

$$-2\left(\frac{\sin 2x}{2}\right)^2 - 1$$

Question [JEE Mains-2023]

The set of all values of λ for which the equation $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$ has a real solution x, is

A [−2, −1]

B [−3/2, −1]

C [−2, −3/2]

D [−1, −1/2]

$$\text{LHS} = -2\frac{\sin^2 2x}{4} - 1$$

$$= -\frac{1}{2}\sin^2 2x - 1.$$

$\sin 2x \in [-1, 1]$

$\sin^2 2x \in [0, 1]$

$-\frac{1}{2}\sin^2 2x \in \left[-\frac{1}{2}, 0\right]$

$-\frac{1}{2}\sin^2 2x - 1 \in \left[-\frac{3}{2}, -1\right] = \lambda$

Question

Find value of λ for which equation $\sin^6 x + \cos^6 x = \lambda - 1$ is consistent.



Question [JEE Mains-2020]

If the equation $\cos^4\theta + \sin^4\theta + \lambda = 0$ has real solutions for θ, then λ lies in the interval



A $[-3/2, -5/2]$

B $[-1/2, -1/4]$

C $[-5/4, -1]$

D $[-1, -1/2]$

DO IT BY YOURSELF (DIBY)

DIBY-01

Solve : $\sqrt{3}\sec 2\theta = 2$.

[Ans. : $\theta = n\pi \pm \frac{\pi}{12}, n \in I$]

DIBY-02

Find general solution of equation $\cot x = \sqrt{3}$

(A) $x = n\pi + \frac{\pi}{3}$

(B) $x = n\pi + \frac{\pi}{6}$

(C) $x = n\pi + \frac{2\pi}{3}$

(D) $x = \frac{9\pi}{6}\frac{5\pi}{6}$

[Ans. : (B)]

DIBY-03

The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is:

(A) 0 (B) 2 (C) 6 (D) 8

[Ans. : (C)]

DIBY-04

General solution of the equation $(2\sin\theta - \sqrt{3})^2 + (\tan\theta + \sqrt{3})^2 = 0$

[Ans. : $(6n+2)\frac{\pi}{3}, n \in I$]

DIBY-05

Find the principal solution of $\sin 2x = -\frac{1}{2}$

(A) $\frac{7\pi}{6}, \frac{11\pi}{6}$

(B) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{15\pi}{12}, \frac{19\pi}{12}$

(C) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$

(D) None of these

[Ans. : (C)]

DIBY-06

Find the number of solution $\tan 3x = 1$ in the interval $x \in \left[-\frac{2\pi}{3}, \frac{5\pi}{6}\right]$.

[Ans. : 05]

DIBY-07

Find the number of solution of the equation $16^{\sin^2 x} + 16^{\cos^2 x} = 10$, where $0 \le x \le 2\pi$.

Ans: 08

DIBY-08

Example 9: Number of values of 'x' in $(-2\pi, 2\pi)$ satisfying the equation $2^{\sin^2 x} + 4(2^{\cos^2 x}) = 6$ is:

(A) 8 (B) 6 (C) 4 (D) 2

Ans: C

DIBY-09

The most general solution of $\tan\theta = -1$ and $\cos\theta = \frac{1}{\sqrt{2}}$ is

(A) $n\pi + \frac{7\pi}{4}, n \in I$

(B) $n\pi + (-1)^n \frac{7\pi}{4}, n \in I$

(C) $2n\pi + \frac{7\pi}{4}, n \in I$

(D) None of these

Ans: C

DIBY-10

If $4\sin^2\theta + 2(\sqrt{3}+1)\cos\theta = 4 + \sqrt{3}$, where $n \in I$ then the general value of θ is

(A) $2n\pi \pm \frac{\pi}{3}$

(B) $2n\pi + \frac{\pi}{4}$

(C) $n\pi \pm \frac{\pi}{3}$

(D) $n\pi - \frac{\pi}{3}$

Ans: A

DIBY-11

Solve for x : $\sin x\left(\cos\frac{x}{4} - 2\sin x\right) + \left(1 + \sin\frac{x}{4} - 2\cos x\right)\cos x = 0$

Ans: $2\pi(4n + 1)$

DIBY-12

The equation $\sqrt{3}\sin x + \cos x = 4$ has

(A) Only one solution

(B) Two solutions

(C) Infinitely many solutions

(D) No solution

Ans: D

DIBY-13

The number of solutions of $\sin\theta + 2\sin 2\theta + 3\sin 3\theta + 4\sin 4\theta = 10$ in $(0, \pi)$ is

Ans: .....

DIBY-14

The general solution of $8\tan^2\frac{x}{2} = 1 + \sec x$ is

(A) $x = 2n\pi \pm \cos^{-1}\left(\frac{-1}{3}\right)$

(B) $x = 2n\pi \pm \frac{\pi}{6}$

(C) $x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right)$

(D) None of these

Ans: C

DIBY-15

Number of solutions of $\sum_{r=1}^{5} \cos rx = 5$ in the interval $[0, 4\pi]$ is:

(A) 0 (B) 2 (C) 3 (D) 7

Ans: C

DIBY-16

$\frac{\sin 3\theta}{2\cos 2\theta + 1} = \frac{1}{2}$ if

(A) $\theta = n\pi + \frac{\pi}{6}, n \in I$

(B) $\theta = 2n\pi - \frac{\pi}{6}, n \in I$

(C) $\theta = n\pi + (-1)^n \frac{\pi}{6}, n \in I$

(D) $\theta = n\pi - \frac{\pi}{6}, n \in I$

Ans: C



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians